

Trader Behaviour Insights under Market Sentiment

1. Objective

The objective of this assignment is to analyze how trader behaviour, performance, and risk exposure change under different market sentiment conditions (Fear, Extreme Fear, Greed, Extreme Greed), and to uncover hidden behavioural patterns that can inform smarter trading strategies.

2. Datasets Overview

2.1 Bitcoin Market Sentiment Dataset

- Provides daily market sentiment labels (Fear, Extreme Fear, Greed, Extreme Greed)
- Used to represent market psychology on each trading day

2.2 Historical Trader Data (Hyperliquid)

- Contains trade-level information such as trader account, coin traded, trade size, execution time, realized PnL, and fees
- Represents actual trader decisions and outcomes

Both datasets were aligned at the daily level using trade execution date.

3. Data Preparation & Methodology

Key preprocessing steps:

- Converted trade timestamps to datetime and extracted trade_date
- Merged trader data with sentiment data on date
- Removed duplicate trades and rows with missing core values (PnL, size, sentiment)
- Derived helper columns:
 - is_win (profit vs loss)
 - pnl_label (Profit / Loss)
 - year (for long-term aggregation)

Analysis was conducted in seven phases, progressing from baseline behaviour to hidden pattern discovery and strategy formulation.

4. Phase-wise Analysis & Key Findings

4.1 Baseline Behaviour vs Sentiment (Phase 2)

Findings

- Trade activity increases significantly during Greed and Extreme Greed
- Average trade size (risk) is highest during Extreme Greed
- Higher activity does not always result in higher average profitability

Interpretation

Traders tend to overtrade and increase exposure during optimistic market conditions.

4.2 Distribution & Risk Analysis (Phase 3)

Findings

- PnL distributions during Extreme Greed show high volatility and large negative outliers
- Fear phases exhibit fewer trades but better loss containment
- Fee accumulation is highest during Greed phases

Interpretation

Emotional trading increases downside risk and fee drag, especially during Extreme Greed.

4.3 Coin-level Sensitivity (Phase 4)

Findings

- Some coins show profitability only during Greed phases
- Certain coins remain stable across all sentiments
- Risk concentration is uneven across coins and sentiments

Interpretation

Not all coins react equally to sentiment; some amplify emotional mistakes.

4.4 Trader-level Consistency (Phase 5)

Findings

- A small subset of traders remains profitable across sentiments
- Many traders profit only in Greed and lose in Fear
- Overtrading traders show high fees with flat or negative PnL

Interpretation

Consistency and risk control distinguish skilled traders from emotional ones.

5. Hidden Pattern Discovery (Phase 6)

The following non-obvious patterns were identified:

1. Risk without reward
 - Risk increases faster than returns during Extreme Greed

2. Overtrading signal
 - High trade count + high fees + low net PnL
3. Sentiment dependency
 - Traders profitable only in Greed are highly sentiment-dependent
4. Consistency vs luck
 - Low PnL volatility correlates with sustainable profitability

These patterns are not visible through averages alone and require cross-metric analysis.

6. Trading Strategy Implications (Phase 7)

Based on the discovered patterns, the following rules are recommended:

- Reduce position size and trade frequency during Extreme Greed
- Favor selective, high-conviction trades during Fear
- Allocate capital to sentiment-agnostic coins
- Scale traders with positive risk-adjusted returns
- Penalize strategies where fees dominate profits

These rules aim to reduce emotional exposure and improve long-term consistency.

7. Conclusion

- Trader behaviour is strongly influenced by market sentiment, with significantly higher trading activity and risk exposure during Greed and Extreme Greed phases.
- Increased risk-taking during Greed does not result in proportional profitability, indicating inefficient and emotionally driven trading behaviour.
- Fear and Neutral market conditions exhibit healthier risk–reward dynamics, with more controlled losses and better risk-adjusted performance despite lower trade volume.
- Correlation analysis shows that larger trade sizes reduce per-trade efficiency, confirming that bigger positions do not guarantee better outcomes.
- Overtrading is a key hidden driver of underperformance, where higher trade counts and fees erode net returns, especially in optimistic market conditions.
- Transaction costs scale with risk and amplify losses during Greed, further weakening overall performance.
- Overall, trader performance is behaviourally dependent on market sentiment, but sustainable profitability depends on risk control and discipline rather than sentiment-driven aggression.

